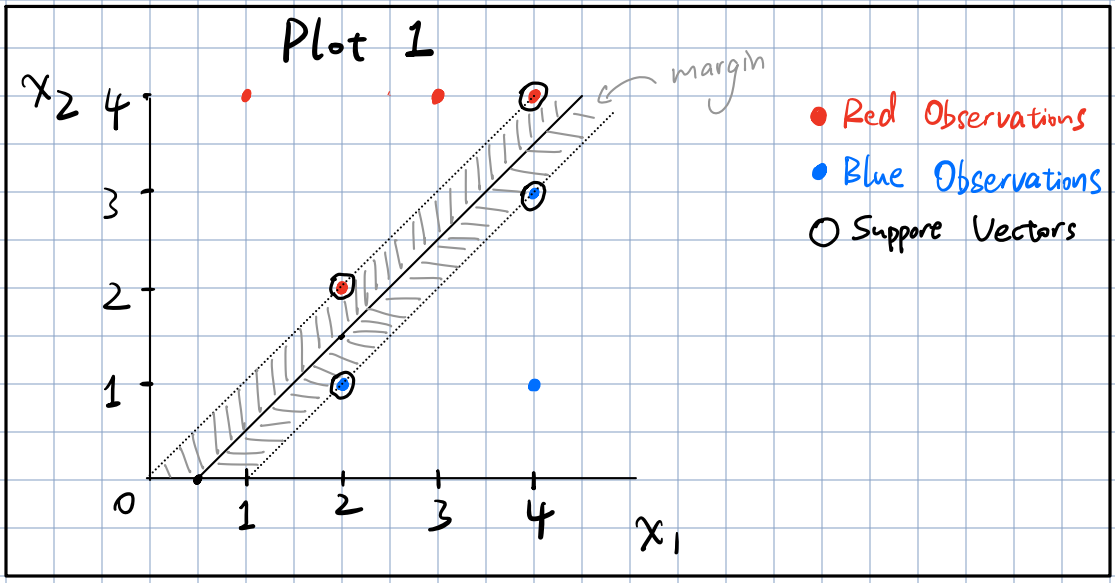


(a) See **Plot 1.**

(b) Hyperplane equation: $\boxed{\frac{1}{2} - X_1 + X_2 = 0}$.

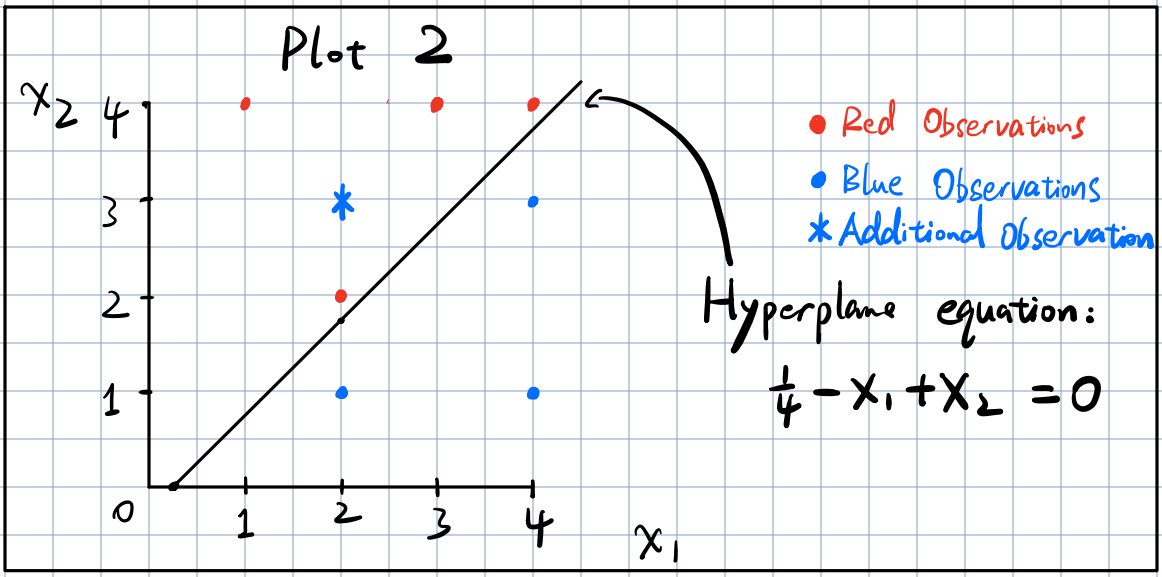
(c) Classify an observation as Red if $\frac{1}{2} - X_1 + X_2 > 0$, Blue otherwise.

(d) The margin for the maximal margin hyperplane is shown in **Plot 1** above as the gray shaded area.

(e) Suppore vectors are $(2,2), (4,4), (2,1), (4,3)$.

(f) Observation 7 has coordinate $(4,1)$, which lies below the Blue margin boundary $X_2 = X_1 - 1$. So it's not a suppore vector. A slight movement will still keep it below that boundary and will not affect the maximal margin hyperplane.

(g)



Although the hyperplane separates observations belonging to different classes, its minimum distance is

$$\frac{1}{4\sqrt{2}} < \frac{1}{2\sqrt{2}}.$$

The margin is smaller than the optimal margin which uses 1(b) equation.

(h) As shown in plot 2, when we add an additional observation $*: (2,3)$, the plot can no longer be separated by a hyperplane.

This is because the point lies inside the triangle formed by the Red points $(1,4), (2,2), (3,4)$. Any half-plane containing these 3 Red points also contains $(2,3)$. So the 2 classes are no longer linearly separable.